Darknet Traffic Detection Project

In [2]:
```
# Import required libraries
import pandas as pd
import numpy as np
```

Phase 1 — Dataset Acquisition

In [3]:
```
# Load the dataset from Excel
# Make sure Darknet.xlsx is in the same folder as this notebook
df = pd.read_excel("Darknet.xlsx")

# Remove empty columns if any exist
df = df.loc[:, ~df.columns.str.contains("^Unnamed")]

print("Dataset shape:", df.shape)
print("Number of columns:", len(df.columns))
```

```
Dataset shape: (141531, 85)
Number of columns: 85
```

In [4]:
```
# View first rows
df.head()
```

Out[4]:

	Flow ID	Src IP	Src Port	Dst IP	Dst Port	Protocol	Timestamp	Fl Durat
0	10.152.152.11-216.58.220.99-57158-443-6	10.152.152.11	57158	216.58.220.99	443	6	24/07/2015 04:09:48 PM	
1	10.152.152.11-216.58.220.99-57159-443-6	10.152.152.11	57159	216.58.220.99	443	6	24/07/2015 04:09:48 PM	
2	10.152.152.11-216.58.220.99-57160-443-6	10.152.152.11	57160	216.58.220.99	443	6	24/07/2015 04:09:48 PM	
3	10.152.152.11-74.125.136.120-49134-443-6	10.152.152.11	49134	74.125.136.120	443	6	24/07/2015 04:09:48 PM	
4	10.152.152.11-173.194.65.127-34697-19305-6	10.152.152.11	34697	173.194.65.127	19305	6	24/07/2015 04:09:45 PM	10778

5 rows × 85 columns

In [5]:
```
# Check dataset information (column types, null counts)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 141531 entries, 0 to 141530
Data columns (total 85 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   Flow ID                     141531 non-null  object
 1   Src IP                      141531 non-null  object
 2   Src Port                    141531 non-null  int64
 3   Dst IP                      141531 non-null  object
 4   Dst Port                    141531 non-null  int64
 5   Protocol                    141531 non-null  int64
 6   Timestamp                   141531 non-null  object
 7   Flow Duration               141531 non-null  int64
 8   Total Fwd Packet            141531 non-null  int64
 9   Total Bwd packets           141531 non-null  int64
 10  Total Length of Fwd Packet  141531 non-null  int64
 11  Total Length of Bwd Packet  141531 non-null  int64
 12  Fwd Packet Length Max       141531 non-null  int64
 13  Fwd Packet Length Min       141531 non-null  int64
 14  Fwd Packet Length Mean      141531 non-null  float64
 15  Fwd Packet Length Std       141531 non-null  float64
 16  Bwd Packet Length Max       141531 non-null  int64
 17  Bwd Packet Length Min       141531 non-null  int64
 18  Bwd Packet Length Mean      141531 non-null  float64
 19  Bwd Packet Length Std       141531 non-null  float64
 20  Flow Bytes/s                141484 non-null  float64
 21  Flow Packets/s              141531 non-null  float64
 22  Flow IAT Mean               141531 non-null  float64
 23  Flow IAT Std                141531 non-null  float64
 24  Flow IAT Max                141531 non-null  int64
 25  Flow IAT Min                141531 non-null  int64
 26  Fwd IAT Total               141531 non-null  int64
 27  Fwd IAT Mean                141531 non-null  float64
 28  Fwd IAT Std                 141531 non-null  float64
 29  Fwd IAT Max                 141531 non-null  int64
 30  Fwd IAT Min                 141531 non-null  int64
 31  Bwd IAT Total               141531 non-null  int64
 32  Bwd IAT Mean                141531 non-null  float64
 33  Bwd IAT Std                 141531 non-null  float64
 34  Bwd IAT Max                 141531 non-null  int64
 35  Bwd IAT Min                 141531 non-null  int64
 36  Fwd PSH Flags               141531 non-null  int64
 37  Bwd PSH Flags               141531 non-null  int64
 38  Fwd URG Flags               141531 non-null  int64
 39  Bwd URG Flags               141531 non-null  int64
 40  Fwd Header Length           141531 non-null  object
 41  Bwd Header Length           141531 non-null  object
 42  Fwd Packets/s               141531 non-null  float64
 43  Bwd Packets/s               141531 non-null  object
 44  Packet Length Min           141531 non-null  int64
 45  Packet Length Max           141531 non-null  int64
 46  Packet Length Mean          141531 non-null  object
 47  Packet Length Std           141531 non-null  float64
 48  Packet Length Variance      141531 non-null  float64
 49  FIN Flag Count              141531 non-null  int64
 50  SYN Flag Count              141531 non-null  int64
```

```
 51  RST Flag Count              141531 non-null  int64
 52  PSH Flag Count              141531 non-null  int64
 53  ACK Flag Count              141531 non-null  int64
 54  URG Flag Count              141531 non-null  int64
 55  CWE Flag Count              141531 non-null  int64
 56  ECE Flag Count              141531 non-null  int64
 57  Down/Up Ratio               141531 non-null  int64
 58  Average Packet Size         141531 non-null  float64
 59  Fwd Segment Size Avg        141531 non-null  float64
 60  Bwd Segment Size Avg        141531 non-null  float64
 61  Fwd Bytes/Bulk Avg          141531 non-null  float64
 62  Fwd Packet/Bulk Avg         141531 non-null  int64
 63  Fwd Bulk Rate Avg           141531 non-null  int64
 64  Bwd Bytes/Bulk Avg          141531 non-null  int64
 65  Bwd Packet/Bulk Avg         141531 non-null  int64
 66  Bwd Bulk Rate Avg           141531 non-null  int64
 67  Subflow Fwd Packets         141531 non-null  int64
 68  Subflow Fwd Bytes           141531 non-null  int64
 69  Subflow Bwd Packets         141531 non-null  int64
 70  Subflow Bwd Bytes           141531 non-null  int64
 71  FWD Init Win Bytes          141531 non-null  int64
 72  Bwd Init Win Bytes          141531 non-null  int64
 73  Fwd Act Data Pkts           141531 non-null  int64
 74  Fwd Seg Size Min            141531 non-null  int64
 75  Active Mean                 141531 non-null  int64
 76  Active Std                  141531 non-null  int64
 77  Active Max                  141531 non-null  int64
 78  Active Min                  141531 non-null  int64
 79  Idle Mean                   141531 non-null  int64
 80  Idle Std                    141531 non-null  float64
 81  Idle Max                    141531 non-null  int64
 82  Idle Min                    141531 non-null  float64
 83  Label                       141531 non-null  object
 84  Label.1                     141531 non-null  object
dtypes: float64(21), int64(54), object(10)
memory usage: 91.8+ MB
```

Phase 2 — Data Preprocessing

In [6]:

```python
# Some columns were loaded as 'object' (text) but should be numbers.
# pd.to_numeric will convert them, and any bad values become NaN.
# ----------------------------------------------------------------
mixed_type_cols = [
    'Fwd Header Length',
    'Bwd Header Length',
    'Bwd Packets/s',
    'Packet Length Mean'
]

for col in mixed_type_cols:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors='coerce')

print("Mixed-type columns converted to numeric.")
```

```
Mixed-type columns converted to numeric.
```

In [7]:
```
# Replace infinite values with NaN, then drop all rows with NaN
df.replace([np.inf, -np.inf], np.nan, inplace=True)
df.dropna(inplace=True)

print('Dataset shape after cleaning NaN/Inf rows:', df.shape)
```

```
Dataset shape after cleaning NaN/Inf rows: (141481, 85)
```

In [8]:
```
# Remove socket-related columns AND Timestamp
# These are identifiers, not useful features for ML
columns_to_remove = [
    'Flow ID',
    'Src IP',
    'Dst IP',
    'Src Port',
    'Dst Port',
    'Timestamp'
]

df.drop(columns=columns_to_remove, inplace=True, errors='ignore')

print('Dataset shape after removing socket/timestamp columns:', df.shape)
print('Remaining columns:', df.shape[1])
```

```
Dataset shape after removing socket/timestamp columns: (141481, 79)
Remaining columns: 79
```

Phase 3 — Label Engineering

The dataset has two label columns:

- `Label` → contains: Tor, Non-Tor, VPN, NonVPN (this is for Case 2)
- `Label.1` → contains: traffic types like P2P, Browsing, etc.

Case 1 (Binary): We need to create a Benign vs Darknet label.

- "Non-Tor" maps to → **Benign**
- "Tor", "VPN", "NonVPN" map to → **Darknet**

Case 2 (Multiclass): Tor / Non-Tor / VPN / Non-VPN

- We use the `Label` column directly, but fix naming inconsistencies.

In [9]:
```
# Some rows have numeric values like '5494.505495' in the Label column
# These are invalid — keep only known valid labels
# ----------------------------------------------------------------
valid_labels = ['Non-Tor', 'Tor', 'VPN', 'NonVPN']
df = df[df['Label'].isin(valid_labels)]

print("Rows after removing corrupted label rows:", len(df))
print("\nLabel column unique values:", df['Label'].unique())
```

```
Rows after removing corrupted label rows: 141481

Label column unique values: ['Non-Tor' 'NonVPN' 'Tor' 'VPN']
```

In [10]:
```
# ----------------------------------------------------------------
# CASE 1: Binary Classification — Benign vs Darknet
# Map: Non-Tor → Benign | Tor, VPN, NonVPN → Darknet
# ----------------------------------------------------------------
label_map_binary = {
    'Non-Tor': 'Benign',
    'Tor':     'Darknet',
    'VPN':     'Darknet',
    'NonVPN':  'Darknet'
}

df['Label_Binary'] = df['Label'].map(label_map_binary)

print("Case 1 — Binary label distribution:")
print(df['Label_Binary'].value_counts())
```

```
Case 1 — Binary label distribution:
Label_Binary
Benign     93309
Darknet    48172
Name: count, dtype: int64
```

In [11]:
```
# ----------------------------------------------------------------
# CASE 2: Multiclass Classification — Tor / Non-Tor / VPN / Non-VPN
# The Label column already has these values, but 'NonVPN' should be
# standardised to 'Non-VPN' to match the paper
# ----------------------------------------------------------------
label_map_multi = {
    'Non-Tor': 'Non-Tor',
    'Tor':     'Tor',
    'VPN':     'VPN',
    'NonVPN':  'Non-VPN'   # standardise naming
}

df['Label_Multi'] = df['Label'].map(label_map_multi)

print("Case 2 — Multiclass label distribution:")
print(df['Label_Multi'].value_counts())
```

```
Case 2 — Multiclass label distribution:
Label_Multi
Non-Tor    93309
Non-VPN    23861
VPN        22919
Tor         1392
Name: count, dtype: int64
```

In [12]:
```
# ----------------------------------------------------------------
# Prepare X (features) and y (labels) for both cases
# Drop all label columns from features
# ----------------------------------------------------------------
label_cols = ['Label', 'Label.1', 'Label_Binary', 'Label_Multi']
```

```
X = df.drop(columns=label_cols, errors='ignore')
y_binary = df['Label_Binary']   # Case 1
y_multi  = df['Label_Multi']    # Case 2

print("Feature matrix shape (X):", X.shape)
print("Binary labels shape   (y_binary):", y_binary.shape)
print("Multi  labels shape   (y_multi):", y_multi.shape)
```

```
Feature matrix shape (X): (141481, 77)
Binary labels shape   (y_binary): (141481,)
Multi  labels shape   (y_multi): (141481,)
```

In [13]:

```
# Final sanity check — make sure X has only numeric columns
non_numeric = X.select_dtypes(exclude=[np.number]).columns.tolist()
if non_numeric:
    print("WARNING: These columns are still non-numeric and will cause problems:",
else:
    print("All feature columns are numeric. Ready for Phase 4 (Train/Test Split)!")

print("\nFinal feature count:", X.shape[1])
print("Final row count:", X.shape[0])
```

```
All feature columns are numeric. Ready for Phase 4 (Train/Test Split)!

Final feature count: 77
Final row count: 141481
```

Phase 4 — Train / Test Split

In [14]:

```
from sklearn.model_selection import train_test_split, StratifiedKFold

# ── Case 1: Binary (Benign vs Darknet) ──────────────────────────────
X_train_bin, X_test_bin, y_train_bin, y_test_bin = train_test_split(
    X, y_binary,
    test_size=0.2,        # 20% goes to test
    random_state=42,      # makes results reproducible
    stratify=y_binary     # keeps class balance (important!)
)

# ── Case 2: Multiclass (Tor / Non-Tor / VPN / Non-VPN) ──────────────
X_train_mul, X_test_mul, y_train_mul, y_test_mul = train_test_split(
    X, y_multi,
    test_size=0.2,
    random_state=42,
    stratify=y_multi
)

# ── 5-Fold Cross-Validation setup (used later during training) ───────
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# ── Confirm shapes ──────────────────────────────────────────────────
print("=== Case 1 — Binary ===")
print(f"  Train: {X_train_bin.shape}  |  Test: {X_test_bin.shape}")
print(f"  Train labels:\n{y_train_bin.value_counts()}")
print(f"\n  Test labels:\n{y_test_bin.value_counts()}")
```

```
print("\n=== Case 2 — Multiclass ===")
print(f"  Train: {X_train_mul.shape}  |  Test: {X_test_mul.shape}")
print(f"  Train labels:\n{y_train_mul.value_counts()}")
```

```
=== Case 1 — Binary ===
  Train: (113184, 77)  |  Test: (28297, 77)
  Train labels:
Label_Binary
Benign     74647
Darknet    38537
Name: count, dtype: int64

  Test labels:
Label_Binary
Benign     18662
Darknet     9635
Name: count, dtype: int64

=== Case 2 — Multiclass ===
  Train: (113184, 77)  |  Test: (28297, 77)
  Train labels:
Label_Multi
Non-Tor    74647
Non-VPN    19089
VPN        18335
Tor         1113
Name: count, dtype: int64
```

Phase 5 — Train ML Models

In [21]:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_score
from sklearn.preprocessing import StandardScaler
import time

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

def make_models():
    """Always returns FRESH, unfitted model instances."""
    return {
        'kNN':           KNeighborsClassifier(n_neighbors=8),
        'MLP':           MLPClassifier(hidden_layer_sizes=(100,), max_iter=300, ran
        'Random Forest': RandomForestClassifier(max_depth=16, n_estimators=50, rand
        'Decision Tree': DecisionTreeClassifier(random_state=42),
        'Grad. Boost':   GradientBoostingClassifier(n_estimators=100, random_state=
    }

# — Scale features for MLP
scaler  = StandardScaler()
scaler2 = StandardScaler()
X_train_bin_scaled = scaler.fit_transform(X_train_bin)
```

```
X_test_bin_scaled  = scaler.transform(X_test_bin)
X_train_mul_scaled = scaler2.fit_transform(X_train_mul)
X_test_mul_scaled  = scaler2.transform(X_test_mul)

def train_models(X_train, y_train,
                 X_train_scaled, case_name):
    print(f"\n{'='*50}")
    print(f"  {case_name}")
    print(f"{'='*50}")
    trained = {}
    for name, model in make_models().items():        # fresh models every time!
        start = time.time()
        X_tr = X_train_scaled if name == 'MLP' else X_train
        cv_scores = cross_val_score(model, X_tr, y_train, cv=cv, scoring='accuracy'
        model.fit(X_tr, y_train)
        elapsed = time.time() - start
        trained[name] = model
        print(f"  {name:<20} CV Accuracy: {cv_scores.mean():.4f} ± {cv_scores.std()
    return trained

print("Training started — this may take a few minutes...")

trained_bin = train_models(X_train_bin, y_train_bin,
                           X_train_bin_scaled,
                           "Case 1 — Binary (Benign vs Darknet)")

trained_mul = train_models(X_train_mul, y_train_mul,
                           X_train_mul_scaled,
                           "Case 2 — Multiclass (Tor/Non-Tor/VPN/Non-VPN)")

print("\nAll models trained successfully!")
```

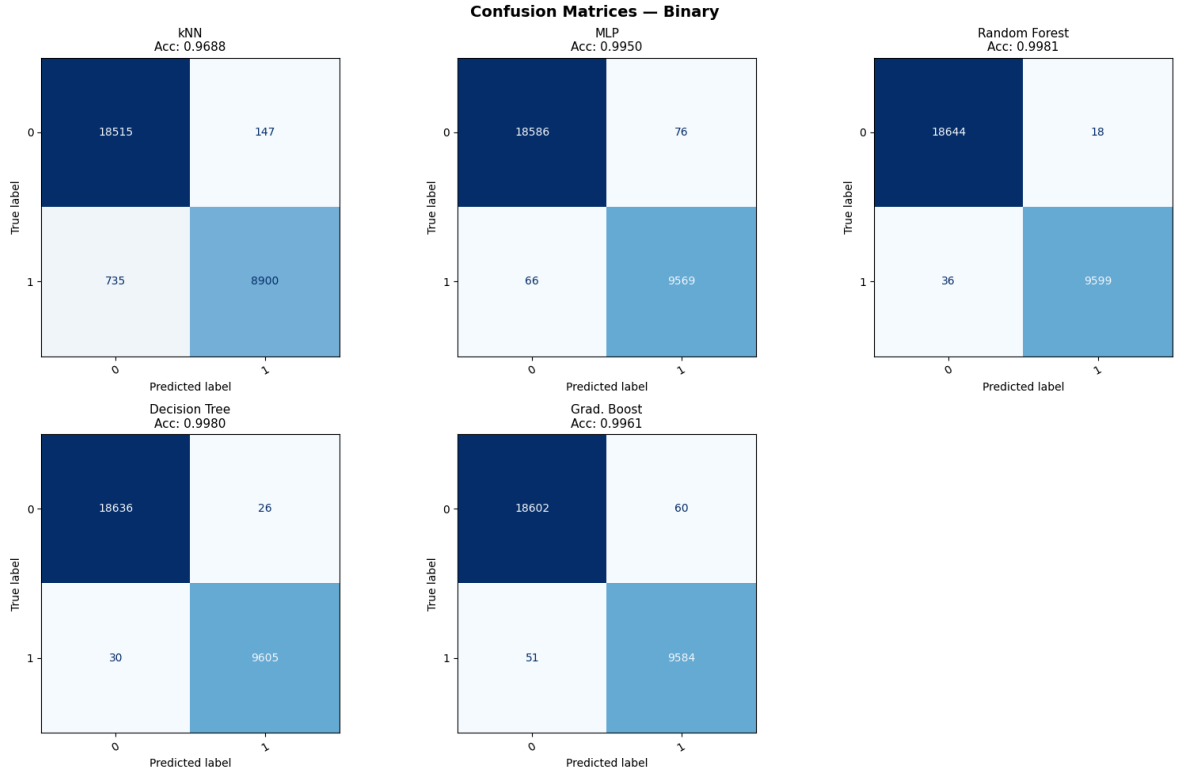
```
Training started — this may take a few minutes...

==================================================
  Case 1 — Binary (Benign vs Darknet)
==================================================
  kNN                  CV Accuracy: 0.9679 ± 0.0008  (7.7s)
  MLP                  CV Accuracy: 0.9943 ± 0.0007  (950.6s)
  Random Forest        CV Accuracy: 0.9982 ± 0.0003  (58.3s)
  Decision Tree        CV Accuracy: 0.9980 ± 0.0003  (16.8s)
  Grad. Boost          CV Accuracy: 0.9966 ± 0.0002  (556.9s)

==================================================
  Case 2 — Multiclass (Tor/Non-Tor/VPN/Non-VPN)
==================================================
  kNN                  CV Accuracy: 0.9515 ± 0.0013  (16.4s)
  MLP                  CV Accuracy: 0.9593 ± 0.0014  (1316.7s)
  Random Forest        CV Accuracy: 0.9833 ± 0.0005  (36.4s)
  Decision Tree        CV Accuracy: 0.9793 ± 0.0009  (7.7s)
  Grad. Boost          CV Accuracy: 0.9769 ± 0.0006  (1161.6s)

All models trained successfully!
```

Phase 6 — Evaluation Metrics

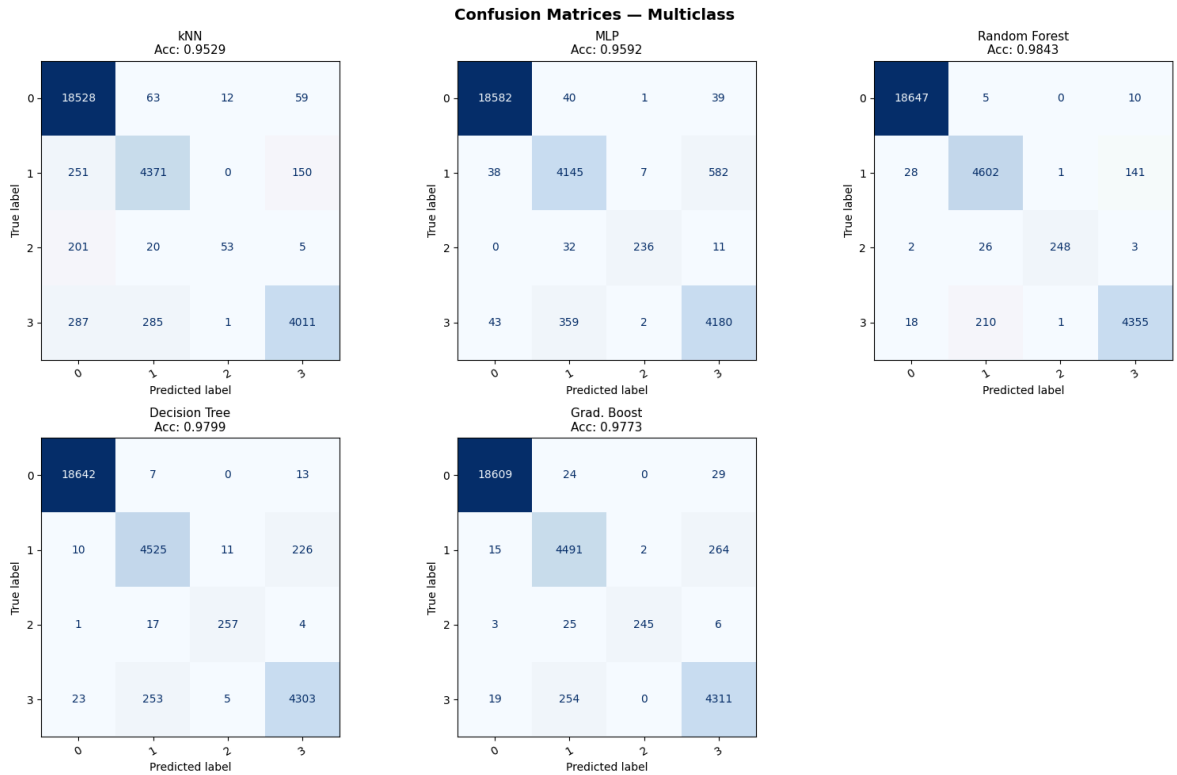
In [22]:
```
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score,
                             confusion_matrix, ConfusionMatrixDisplay)
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

def evaluate_models(trained, case_num, X_test, y_test, case_name):
    print(f"\n{'='*65}")
    print(f"  {case_name}")
    print(f"{'='*65}")
    print(f"  {'Model':<20} {'Accuracy':>9} {'Precision':>10} {'Recall':>8} {'F-sco
    print(f"  {'-'*58}")

    results = {}
    for name, model in trained.items():
        X_eval = (X_test_bin_scaled if case_num == 1 else X_test_mul_scaled) \
                 if name == 'MLP' else X_test

        y_pred = model.predict(X_eval)
        acc  = accuracy_score(y_test, y_pred)
        prec = precision_score(y_test, y_pred, average='weighted', zero_division=0)
        rec  = recall_score(y_test, y_pred, average='weighted', zero_division=0)
        f1   = f1_score(y_test, y_pred, average='weighted', zero_division=0)

        results[name] = {'accuracy': acc, 'precision': prec,
                         'recall': rec,  'f1': f1, 'y_pred': y_pred}
        print(f"  {name:<20} {acc:>8.4f}  {prec:>9.4f}  {rec:>7.4f}  {f1:>8.4f}")

    return results

results_bin = evaluate_models(trained_bin, 1, X_test_bin, y_test_bin,
                              "Case 1 — Binary (Benign vs Darknet)")
results_mul = evaluate_models(trained_mul, 2, X_test_mul, y_test_mul,
                              "Case 2 — Multiclass (Tor/Non-Tor/VPN/Non-VPN)")

# ── Confusion Matrices
def plot_confusion_matrices(trained, results, y_test, case_name):
    fig, axes = plt.subplots(2, 3, figsize=(16, 10))
    axes = axes.flatten()
    fig.suptitle(f'Confusion Matrices — {case_name}', fontsize=14, fontweight='bold

    for i, (name, _) in enumerate(trained.items()):
        y_pred = results[name]['y_pred']
        cm = confusion_matrix(y_test, y_pred)
        disp = ConfusionMatrixDisplay(confusion_matrix=cm)
        disp.plot(ax=axes[i], colorbar=False, cmap='Blues')
        axes[i].set_title(f'{name}\nAcc: {results[name]["accuracy"]:.4f}', fontsize
        axes[i].tick_params(axis='x', rotation=30)

    axes[5].axis('off')
    plt.tight_layout()
    fname = 'confusion_matrices_binary.png' if 'Binary' in case_name else 'confusio
    plt.savefig(fname, dpi=150, bbox_inches='tight')
    plt.show()
```

```
    print(f"Saved: {fname}")

plot_confusion_matrices(trained_bin, results_bin, y_test_bin, "Binary")
plot_confusion_matrices(trained_mul, results_mul, y_test_mul, "Multiclass")

print("\nPhase 6 complete!")
```

```
================================================================
  Case 1 — Binary (Benign vs Darknet)
================================================================
  Model                 Accuracy  Precision   Recall   F-score
  -------------------------------------------------------
  kNN                    0.9688     0.9693   0.9688    0.9686
  MLP                    0.9950     0.9950   0.9950    0.9950
  Random Forest          0.9981     0.9981   0.9981    0.9981
  Decision Tree          0.9980     0.9980   0.9980    0.9980
  Grad. Boost            0.9961     0.9961   0.9961    0.9961

================================================================
  Case 2 — Multiclass (Tor/Non-Tor/VPN/Non-VPN)
================================================================
  Model                 Accuracy  Precision   Recall   F-score
  -------------------------------------------------------
  kNN                    0.9529     0.9515   0.9529    0.9499
  MLP                    0.9592     0.9596   0.9592    0.9592
  Random Forest          0.9843     0.9843   0.9843    0.9842
  Decision Tree          0.9799     0.9798   0.9799    0.9798
  Grad. Boost            0.9773     0.9774   0.9773    0.9774
```




```
Saved: confusion_matrices_binary.png
```



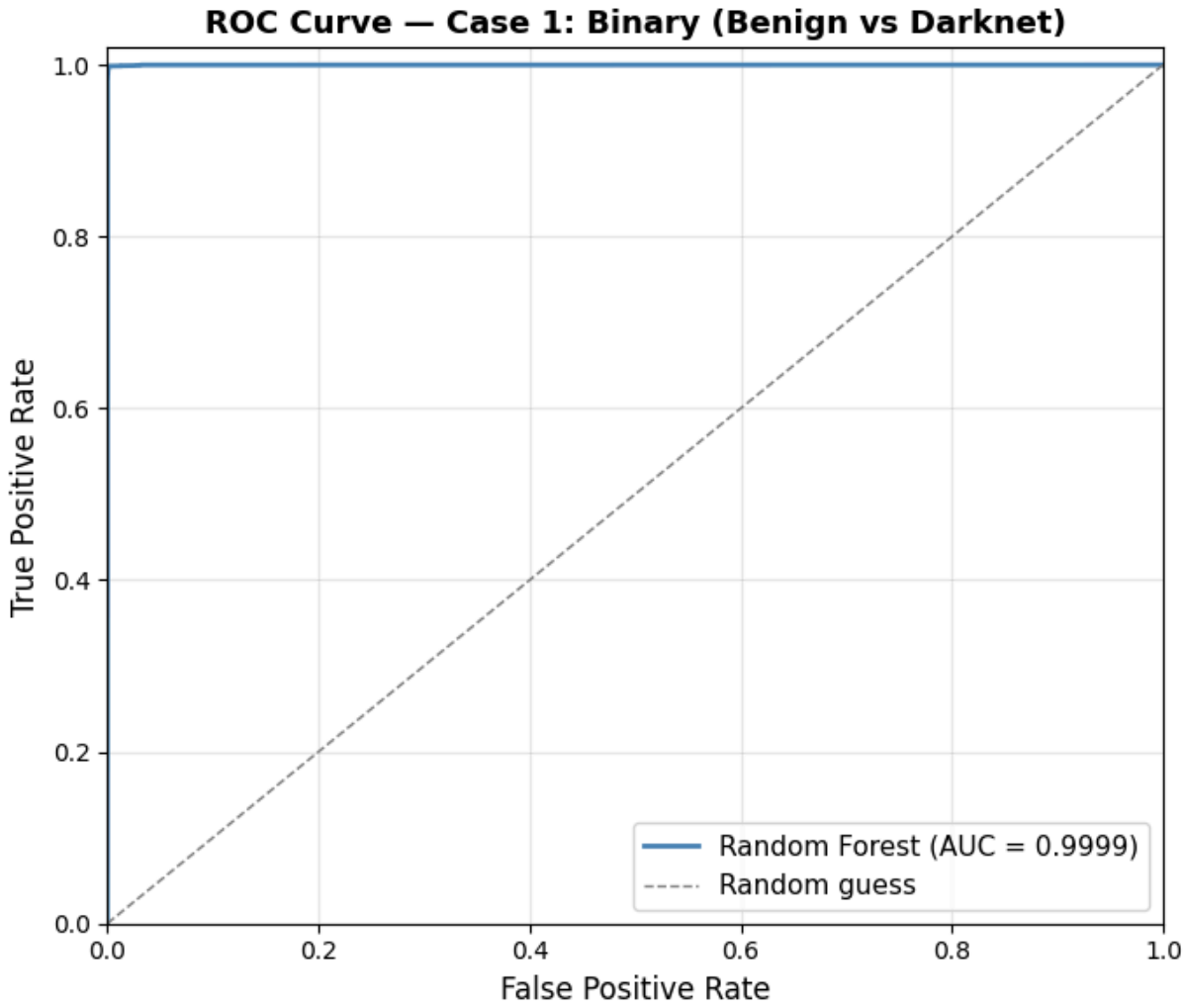

```
Saved: confusion_matrices_multi.png

Phase 6 complete!
```

PHASE 7 — ROC Analysis (Random Forest — Best Classifier)

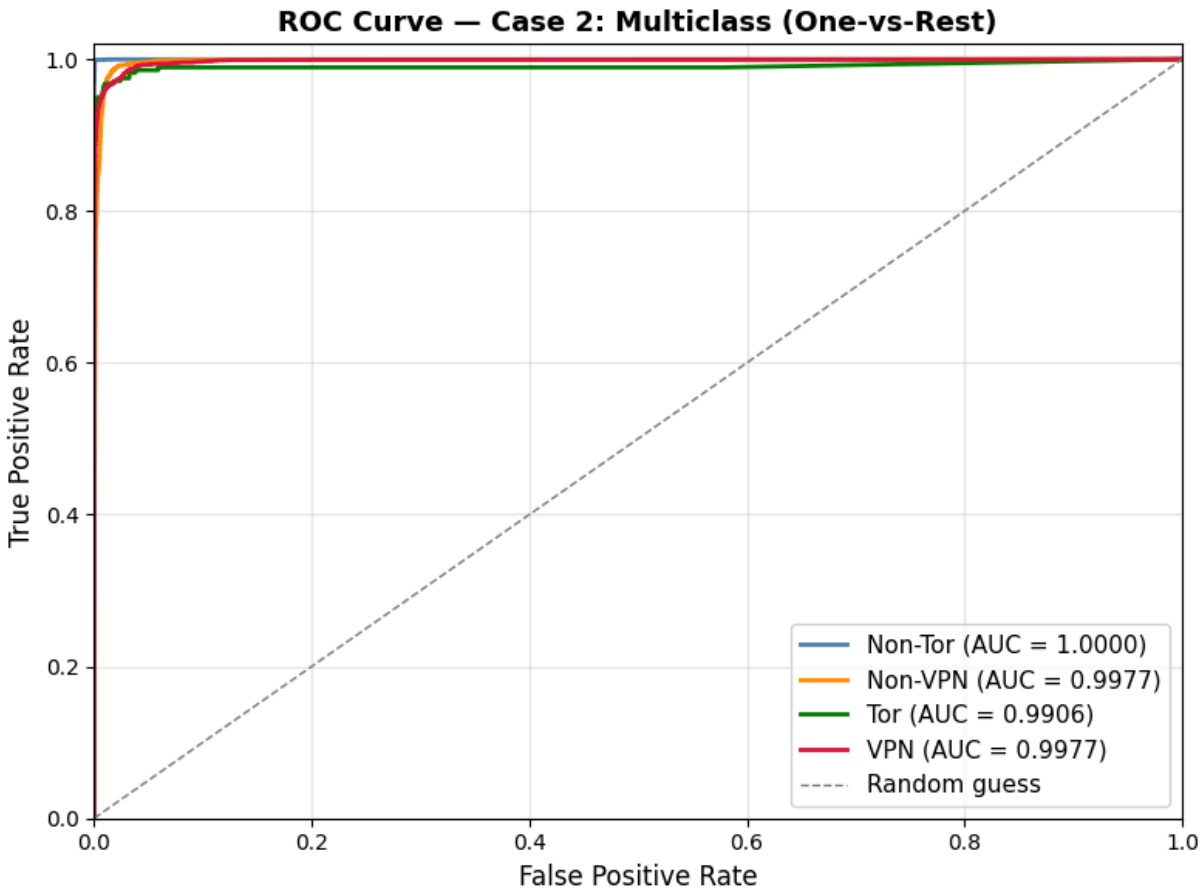
In [23]:

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

rf_bin = trained_bin['Random Forest']
rf_mul = trained_mul['Random Forest']

# CASE 1: Binary ROC
fig, ax = plt.subplots(figsize=(7, 6))

y_score_bin = rf_bin.predict_proba(X_test_bin)[:, 1]

# Map labels to 0/1 for roc_curve
y_test_bin_num = (y_test_bin == 'Darknet').astype(int)

fpr, tpr, _ = roc_curve(y_test_bin_num, y_score_bin)
roc_auc = auc(fpr, tpr)

ax.plot(fpr, tpr, color='steelblue', lw=2,
        label=f'Random Forest (AUC = {roc_auc:.4f})')
ax.plot([0, 1], [0, 1], color='gray', lw=1, linestyle='--', label='Random guess')
ax.set_xlim([0.0, 1.0])
ax.set_ylim([0.0, 1.02])
```

```
ax.set_xlabel('False Positive Rate', fontsize=12)
ax.set_ylabel('True Positive Rate', fontsize=12)
ax.set_title('ROC Curve — Case 1: Binary (Benign vs Darknet)', fontsize=13, fontwei
ax.legend(loc='lower right', fontsize=11)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('roc_case1_binary.png', dpi=150, bbox_inches='tight')
plt.show()
print(f"Case 1 AUC: {roc_auc:.4f}  |  Saved: roc_case1_binary.png")

# CASE 2: Multiclass ROC (One-vs-Rest)
classes = ['Non-Tor', 'Non-VPN', 'Tor', 'VPN']
colors  = ['steelblue', 'darkorange', 'green', 'crimson']

# Binarize labels for One-vs-Rest
y_test_bin2 = label_binarize(y_test_mul, classes=classes)
y_score_mul = rf_mul.predict_proba(X_test_mul)

# Make sure column order matches 'classes'
class_order = list(rf_mul.classes_)
col_idx = [class_order.index(c) for c in classes]
y_score_mul = y_score_mul[:, col_idx]

fig, ax = plt.subplots(figsize=(8, 6))

for i, (cls, color) in enumerate(zip(classes, colors)):
    fpr_i, tpr_i, _ = roc_curve(y_test_bin2[:, i], y_score_mul[:, i])
    auc_i = auc(fpr_i, tpr_i)
    ax.plot(fpr_i, tpr_i, color=color, lw=2,
            label=f'{cls} (AUC = {auc_i:.4f})')

ax.plot([0, 1], [0, 1], color='gray', lw=1, linestyle='--', label='Random guess')
ax.set_xlim([0.0, 1.0])
ax.set_ylim([0.0, 1.02])
ax.set_xlabel('False Positive Rate', fontsize=12)
ax.set_ylabel('True Positive Rate', fontsize=12)
ax.set_title('ROC Curve — Case 2: Multiclass (One-vs-Rest)', fontsize=13, fontweigh
ax.legend(loc='lower right', fontsize=11)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('roc_case2_multiclass.png', dpi=150, bbox_inches='tight')
plt.show()
print("Case 2 AUC per class:")
for i, cls in enumerate(classes):
    fpr_i, tpr_i, _ = roc_curve(y_test_bin2[:, i], y_score_mul[:, i])
    print(f"  {cls:<10} AUC = {auc(fpr_i, tpr_i):.4f}")
print("Saved: roc_case2_multiclass.png")

print("\nPhase 7 complete!")
```




```
Case 1 AUC: 0.9999  |  Saved: roc_case1_binary.png
```



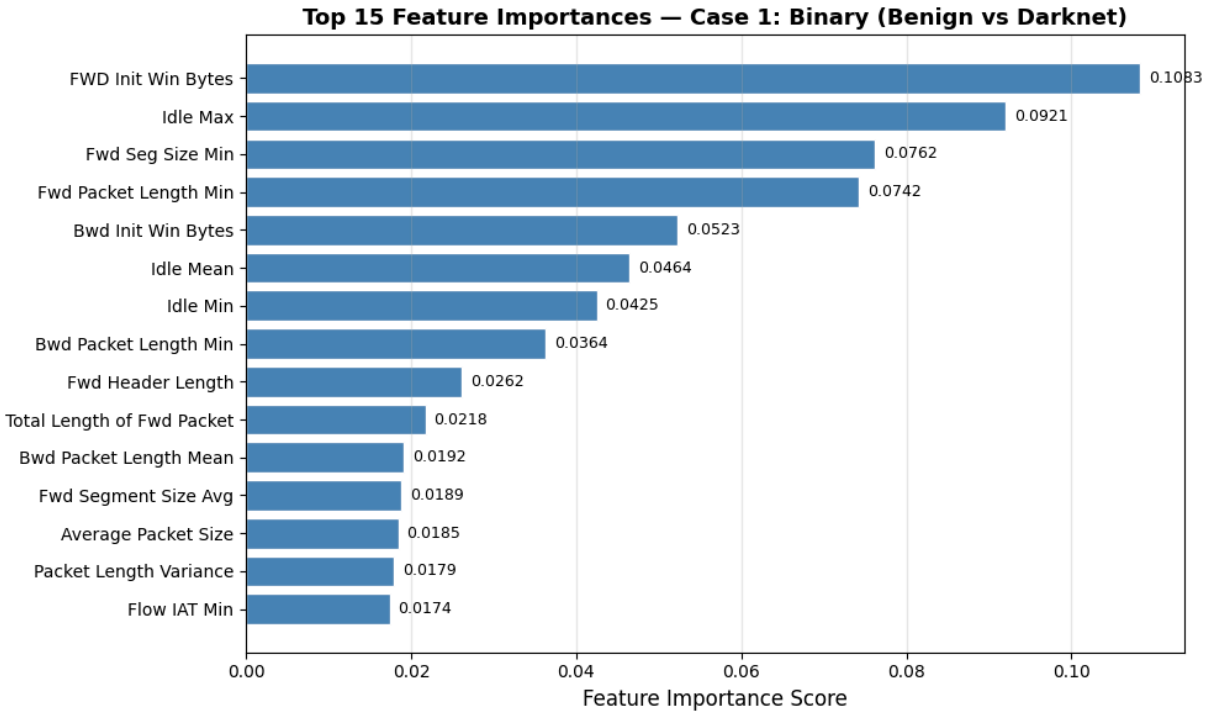

```
Case 2 AUC per class:
  Non-Tor    AUC = 1.0000
  Non-VPN    AUC = 0.9977
  Tor        AUC = 0.9906
  VPN        AUC = 0.9977
Saved: roc_case2_multiclass.png

Phase 7 complete!
```

PHASE 8 — Feature Importance (Random Forest)

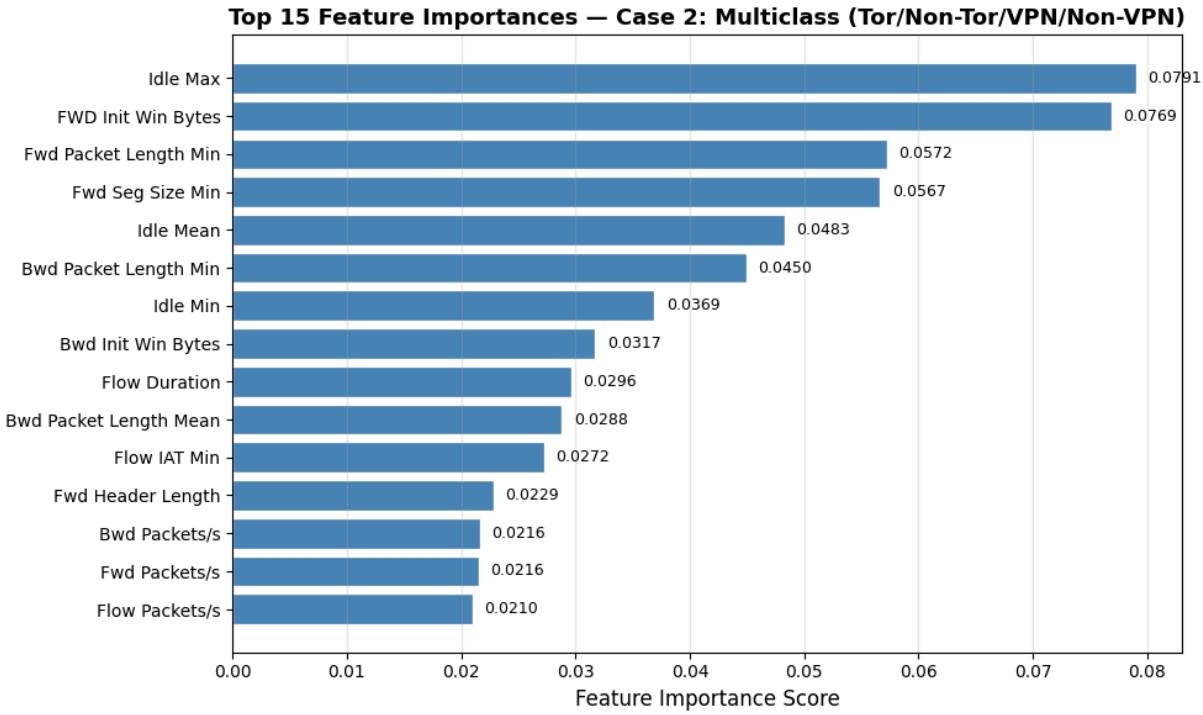
In [24]:

```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

rf_bin = trained_bin['Random Forest']
rf_mul = trained_mul['Random Forest']

feature_names = X_train_bin.columns.tolist()

def plot_feature_importance(model, feature_names, case_name, filename, top_n=15):
    importances = model.feature_importances_
    indices = np.argsort(importances)[::-1][:top_n]

    top_features = [feature_names[i] for i in indices]
    top_values   = importances[indices]

    fig, ax = plt.subplots(figsize=(10, 6))
```

```
    bars = ax.barh(top_features[::-1], top_values[::-1],
                   color='steelblue', edgecolor='white')

    # Add value labels on bars
    for bar, val in zip(bars, top_values[::-1]):
        ax.text(bar.get_width() + 0.001, bar.get_y() + bar.get_height()/2,
                f'{val:.4f}', va='center', fontsize=9)

    ax.set_xlabel('Feature Importance Score', fontsize=12)
    ax.set_title(f'Top {top_n} Feature Importances — {case_name}',
                 fontsize=13, fontweight='bold')
    ax.grid(axis='x', alpha=0.3)
    plt.tight_layout()
    plt.savefig(filename, dpi=150, bbox_inches='tight')
    plt.show()
    print(f"Saved: {filename}")

    # Print ranked list
    print(f"\nTop {top_n} features for {case_name}:")
    for rank, (feat, val) in enumerate(zip(top_features, top_values), 1):
        print(f"  {rank:>2}. {feat:<35} {val:.4f}")

    return top_features, top_values

#  Case 1: Binary
top_feat_bin, top_val_bin = plot_feature_importance(
    rf_bin, feature_names,
    "Case 1: Binary (Benign vs Darknet)",
    "feature_importance_case1.png"
)

#  Case 2: Multiclass
top_feat_mul, top_val_mul = plot_feature_importance(
    rf_mul, feature_names,
    "Case 2: Multiclass (Tor/Non-Tor/VPN/Non-VPN)",
    "feature_importance_case2.png"
)

# Side-by-side comparison of top 5
print("\n" + "="*55)
print("  Top 5 Feature Comparison")
print("="*55)
print(f"  {'Rank':<6} {'Case 1':<35} {'Case 2'}")
print(f"  {'-'*53}")
for i in range(5):
    print(f"  {i+1:<6} {top_feat_bin[i]:<35} {top_feat_mul[i]}")

print("\nPhase 8 complete!")
```




```
Saved: feature_importance_case1.png

Top 15 features for Case 1: Binary (Benign vs Darknet):
   1. FWD Init Win Bytes                 0.1083
   2. Idle Max                           0.0921
   3. Fwd Seg Size Min                   0.0762
   4. Fwd Packet Length Min              0.0742
   5. Bwd Init Win Bytes                 0.0523
   6. Idle Mean                          0.0464
   7. Idle Min                           0.0425
   8. Bwd Packet Length Min              0.0364
   9. Fwd Header Length                  0.0262
  10. Total Length of Fwd Packet         0.0218
  11. Bwd Packet Length Mean             0.0192
  12. Fwd Segment Size Avg               0.0189
  13. Average Packet Size                0.0185
  14. Packet Length Variance             0.0179
  15. Flow IAT Min                       0.0174
```



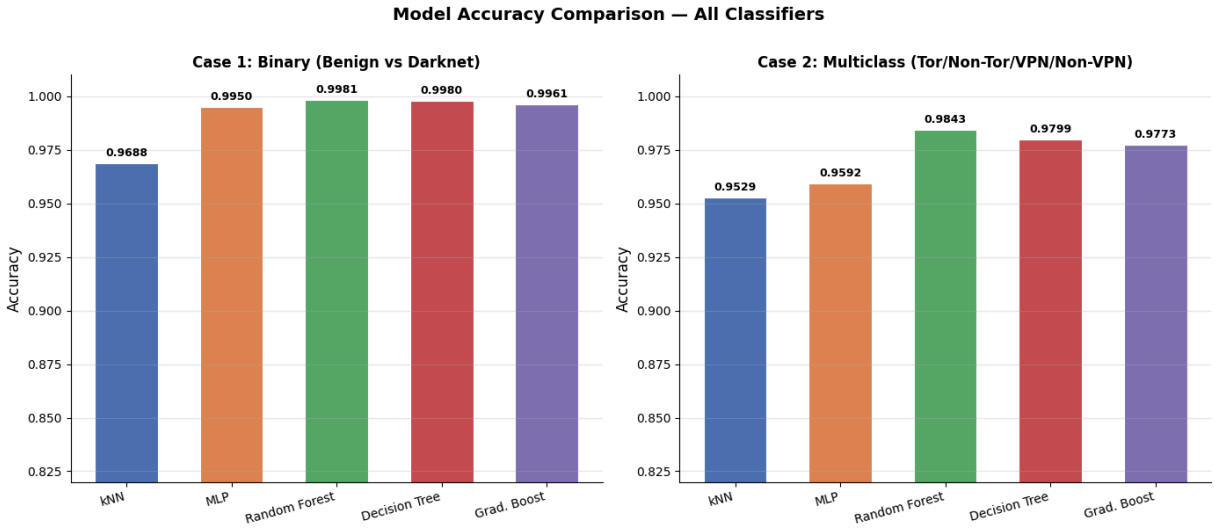

```
Saved: feature_importance_case2.png

Top 15 features for Case 2: Multiclass (Tor/Non-Tor/VPN/Non-VPN):
   1. Idle Max                          0.0791
   2. FWD Init Win Bytes                0.0769
   3. Fwd Packet Length Min             0.0572
   4. Fwd Seg Size Min                  0.0567
   5. Idle Mean                         0.0483
   6. Bwd Packet Length Min             0.0450
   7. Idle Min                          0.0369
   8. Bwd Init Win Bytes                0.0317
   9. Flow Duration                     0.0296
  10. Bwd Packet Length Mean            0.0288
  11. Flow IAT Min                      0.0272
  12. Fwd Header Length                 0.0229
  13. Bwd Packets/s                     0.0216
  14. Fwd Packets/s                     0.0216
  15. Flow Packets/s                    0.0210

=======================================================
  Top 5 Feature Comparison
=======================================================
  Rank   Case 1                              Case 2
  -------------------------------------------------------
  1      FWD Init Win Bytes                  Idle Max
  2      Idle Max                            FWD Init Win Bytes
  3      Fwd Seg Size Min                    Fwd Packet Length Min
  4      Fwd Packet Length Min               Fwd Seg Size Min
  5      Bwd Init Win Bytes                  Idle Mean

Phase 8 complete!
```

Compare and Report Results

In [26]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Build summary DataFrames from Phase 6 results
def build_summary(results, case_name):
    rows = []
    for name, metrics in results.items():
        rows.append({
            'Model':     name,
            'Accuracy':  round(metrics['accuracy'],  4),
            'Precision': round(metrics['precision'], 4),
            'Recall':    round(metrics['recall'],    4),
            'F-score':   round(metrics['f1'],        4),
        })
    df = pd.DataFrame(rows).set_index('Model')
    print(f"\n{'='*60}")
    print(f"  {case_name}")
    print(f"{'='*60}")
    print(df.to_string())
    return df

df_bin = build_summary(results_bin, "Case 1 — Binary (Benign vs Darknet)")
df_mul = build_summary(results_mul, "Case 2 — Multiclass (Tor/Non-Tor/VPN/Non-VPN)"

#  Bar chart: Accuracy comparison
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
colors = ['#4C72B0', '#DD8452', '#55A868', '#C44E52', '#8172B2']
models = list(results_bin.keys())
x = np.arange(len(models))
width = 0.6

for ax, df, title in zip(axes,
                         [df_bin, df_mul],
                         ["Case 1: Binary (Benign vs Darknet)",
                          "Case 2: Multiclass (Tor/Non-Tor/VPN/Non-VPN)"]):
    bars = ax.bar(x, df['Accuracy'], width=width,
                  color=colors, edgecolor='white', linewidth=0.8)

    # Value labels on top of bars
    for bar in bars:
        ax.text(bar.get_x() + bar.get_width()/2,
                bar.get_height() + 0.002,
                f'{bar.get_height():.4f}',
                ha='center', va='bottom', fontsize=9, fontweight='bold')

    ax.set_xticks(x)
    ax.set_xticklabels(df.index, rotation=15, ha='right', fontsize=10)
    ax.set_ylim([0.82, 1.01])
    ax.set_ylabel('Accuracy', fontsize=12)
    ax.set_title(title, fontsize=12, fontweight='bold')
    ax.grid(axis='y', alpha=0.3)
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)
```

```
fig.suptitle('Model Accuracy Comparison — All Classifiers',
             fontsize=14, fontweight='bold', y=1.01)
plt.tight_layout()
plt.savefig('phase9_accuracy_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: phase9_accuracy_comparison.png")

#  Best model summary
best_bin = df_bin['Accuracy'].idxmax()
best_mul = df_mul['Accuracy'].idxmax()

print(f"""
{'='*60}
  FINAL SUMMARY
{'='*60}
  Case 1 Best Model : {best_bin}
    Accuracy  : {df_bin.loc[best_bin, 'Accuracy']:.4f}
    Precision : {df_bin.loc[best_bin, 'Precision']:.4f}
    Recall    : {df_bin.loc[best_bin, 'Recall']:.4f}
    F-score   : {df_bin.loc[best_bin, 'F-score']:.4f}

  Case 2 Best Model : {best_mul}
    Accuracy  : {df_mul.loc[best_mul, 'Accuracy']:.4f}
    Precision : {df_mul.loc[best_mul, 'Precision']:.4f}
    Recall    : {df_mul.loc[best_mul, 'Recall']:.4f}
    F-score   : {df_mul.loc[best_mul, 'F-score']:.4f}

  ROC AUC (Random Forest):
    Case 1 (Binary)     : 0.9999
    Case 2 Non-Tor      : 1.0000
    Case 2 Non-VPN      : 0.9977
    Case 2 VPN          : 0.9977
    Case 2 Tor          : 0.9906

  Top Feature (Case 1)  : FWD Init Win Bytes  (0.1083)
  Top Feature (Case 2)  : Idle Max            (0.0791)
{'='*60}
""")

print("Phase 9 complete! ")
```

```
============================================================
  Case 1 — Binary (Benign vs Darknet)
============================================================
               Accuracy  Precision  Recall  F-score
Model
kNN              0.9688     0.9693  0.9688   0.9686
MLP              0.9950     0.9950  0.9950   0.9950
Random Forest    0.9981     0.9981  0.9981   0.9981
Decision Tree    0.9980     0.9980  0.9980   0.9980
Grad. Boost      0.9961     0.9961  0.9961   0.9961

============================================================
  Case 2 — Multiclass (Tor/Non-Tor/VPN/Non-VPN)
============================================================
               Accuracy  Precision  Recall  F-score
Model
kNN              0.9529     0.9515  0.9529   0.9499
MLP              0.9592     0.9596  0.9592   0.9592
Random Forest    0.9843     0.9843  0.9843   0.9842
Decision Tree    0.9799     0.9798  0.9799   0.9798
Grad. Boost      0.9773     0.9774  0.9773   0.9774
```



```
Saved: phase9_accuracy_comparison.png

============================================================
  FINAL SUMMARY
============================================================
  Case 1 Best Model : Random Forest
    Accuracy  : 0.9981
    Precision : 0.9981
    Recall    : 0.9981
    F-score   : 0.9981

  Case 2 Best Model : Random Forest
    Accuracy  : 0.9843
    Precision : 0.9843
    Recall    : 0.9843
    F-score   : 0.9842

  ROC AUC (Random Forest):
    Case 1 (Binary)     : 0.9999
    Case 2 Non-Tor      : 1.0000
    Case 2 Non-VPN      : 0.9977
    Case 2 VPN          : 0.9977
    Case 2 Tor          : 0.9906

  Top Feature (Case 1)  : FWD Init Win Bytes  (0.1083)
  Top Feature (Case 2)  : Idle Max            (0.0791)
============================================================

Phase 9 complete!
```